

# Absorbing phase transition in a queueing model of coupled adaptive agents

Alexei Vazquez<sup>1,\*</sup>

<sup>1</sup>*Nodes & Links Ltd, Salisbury House, Station Road, Cambridge, CB1 2LA, UK*

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What decides whether people do things together or separately? Many activities cannot be carried out alone, and an individual must rank them against the private tasks competing for the same time. We address this within the priority-queue description of human activity by letting each agent *choose* the priority of a shared task rather than drawing it from a fixed distribution: the value of the joint activity, discounted by the estimated risk that the partner will not take part. Participation becomes strategic, and the model acquires a phase transition. A coupled phase, in which joint activity is sustained, is separated from an absorbing solitary phase by a saddle-node bifurcation that we obtain in closed form. The transition is discontinuous and the solitary phase is absorbing, so collapse is irreversible unless an agent persists unilaterally for of order one memory time, a cost we also compute. The heavy-tailed interevent statistics that motivate queueing models of human dynamics survive only in a narrow window at the transition, and there the exponent is fixed by the fraction of time spent coupled rather than by queue length: the universality classes of the non-strategic model do not survive the introduction of choice. On a network, attention divides as  $1/(k + a)$  and fixes a critical degree beyond which no coupled state exists, so the solitary phase percolates according to the Molloy–Reed criterion with the second moment truncated at that degree — formally an attack on hubs, with no attacker. For group activities the critical degree falls as the  $(m - 1)$ th root, implying a maximum group size. The transition organises two quantities already measured in communication records: a finite capacity for keeping ties active, and the decay of ties whose rhythm is interrupted.

## I. INTRODUCTION

Queueing theory has become a standard framework for the timing of human activity [1–3]. Representing an individual’s to-do list as a priority queue with highest-priority-first execution produces power-law interevent time distributions  $P(\tau) \sim \tau^{-\alpha}$ , with  $\alpha = 1$  and  $\alpha = 3/2$  identified as universality classes and corroborated by electronic and postal correspondence data [1, 3, 4], the  $\alpha = 3/2$  class going back to the single-server analysis of Cobham [7].

Single-queue models describe an isolated individual. Many activities, however, require two or more people at once, and interaction is known from the theory of critical phenomena to be decisive for universality [6]. A minimal two-agent extension [5] gives each agent one *interacting* task  $I$ , executable only when both agents select it in the same step, and one aggregate *non-interacting* task  $O$  standing for  $L_j - 1$  private tasks. That model predicts a countable family of exponents,  $\alpha = 1 + 1/\max_j(L_j - 1)$ , interpolating between 2 and 1: the more distracted agent sets the exponent.

In that treatment the priority of the interacting task is drawn independently from a fixed density. This is the natural assumption if the shared activity is exogenous, but it removes the feature that makes joint tasks distinctive: an agent who is repeatedly stood up should stop offering. Here we make the interacting-task priority a

decision variable. The agents estimate each other’s participation rate from observation and set their priority accordingly. This changes the object from a stochastic process into a coupled inference problem and, as we show, changes its phenomenology substantially.

Strategic queueing is of course an established subject, from Naor [14] onward, but it concerns customers deciding whether to join, balk, or purchase priority in a service queue. What is at issue here is different: the priority of a task that cannot be executed unilaterally, so that the queueing model is simultaneously a coordination game with an absorbing failure state. The resulting phenomenology connects to two literatures that have so far had little contact with the queueing account of human dynamics. The capacity constraint we derive below has been measured directly: mobile-phone records show that individuals possess a finite communication capacity limiting the number of ties they can hold active, and that people with larger networks do not communicate proportionally more but maintain weaker ties [15, 16], while the distribution of effort across alters persists even as the alters themselves turn over [17]. The timescale condition we obtain has likewise been measured: ties whose communication is interrupted for much longer than their established rhythm decay, and temporal features predict tie persistence better than intensity or topology [18]. On the network side, the percolation criterion we derive is formally that of intentional attack [19–21], with the difference that no adversary is present: attention scarcity removes high-degree relations on its own.

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\* alexei@nodeslinks.com

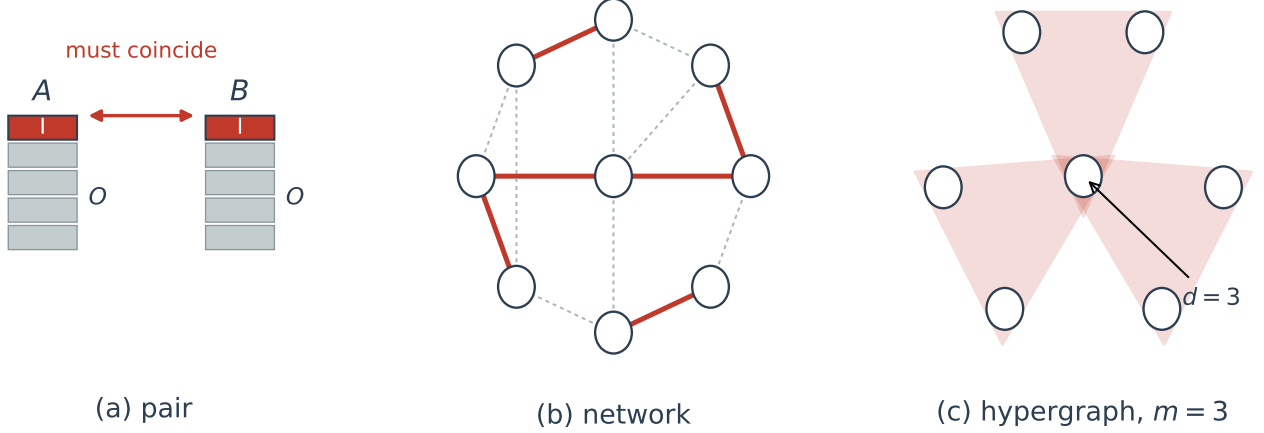


FIG. 1. The three settings. In each, an agent holds one interacting task per shared activity together with an aggregate private task  $O$  standing for  $L - 1$  individual tasks, and executes the single highest-priority item per step. (a) A pair: the shared task  $I$  executes only if both agents select it in the same step. (b) A network: an agent holds one interacting task per neighbour, so attention must be divided; solid links are relations in the coupled phase, dashed ones have collapsed to the solitary phase. (c) A hypergraph: an activity of size  $m$  executes only if all  $m$  members select it, and an agent may belong to  $d$  such groups.

## II. MODEL

### A. Substrate

We retain the structure of Ref. [5]. Agent  $j \in \{A, B\}$  holds one interacting task  $I$  and one aggregate task  $O$  representing  $a_j \equiv L_j - 1$  private tasks. The private-task priority is the maximum of  $a_j$  independent uniform variates,

$$f_{O_j}(x) = a_j x^{a_j-1}, \quad x \in [0, 1], \quad (1)$$

redrawn whenever  $O$  is executed. Each agent selects the higher-priority task.  $I$  executes only if both agents select it.

### B. Priority as a decision

Let  $x_{I_j}$  be the priority agent  $j$  assigns to the interacting task. Since  $O$  is drawn afresh each step, the probability that  $j$  selects  $I$  is

$$p_j = \Pr(x_{O_j} < x_{I_j}) = x_{I_j}^{a_j}. \quad (2)$$

The agent controls a propensity, not a decision: with many private tasks,  $a_j$  large, even a high  $x_{I_j}$  yields a small participation probability.

Payoffs per step: both agents select  $I$ , each receives  $R_I$ ; an agent selects  $I$  while its partner does not, it receives  $x_{O_j} - c$ , the step being wasted and  $c > 0$  the cost of an unreciprocated offer; otherwise it receives  $x_{O_j}$ , the realised value of its private task. The cost  $c$  is what

makes a model of the partner necessary; at  $c = 0$  the optimal policy is  $x_{I_j} = 1$  regardless of the partner.

Writing  $\hat{p}$  for the agent's estimate of its partner's participation probability, selecting  $I$  is preferable whenever  $\hat{p}R_I + (1 - \hat{p})(x_{O_j} - c) > x_{O_j}$ , giving the optimal priority

$$x_{I_j}^* = \left[ R_I - c \frac{1 - \hat{p}}{\hat{p}} \right]_0^1, \quad (3)$$

where  $[\cdot]_0^1$  denotes clipping to  $[0, 1]$ : the value of the interaction discounted by the risk that the partner will not participate. Estimates are maintained by an exponential moving average with retention  $\lambda = e^{-1/\tau_{\text{mem}}}$  over the partner's observed selections.

### C. Levels of modelling

We distinguish three policies by what each agent represents.

**Level 0:** No partner model:  $x_I = R_I$ , clipped.

**Level 1:** Partner model:  $\hat{p}$  tracked, Eq. (3) applied.

**Level 2:** Self model: the agent additionally tracks  $\hat{s}$ , its estimate of the partner's estimate of itself, and evaluates a finite-horizon rollout of the coupled dynamics in which its own participation raises  $\hat{s}$ , which raises the partner's priority through Eq. (3), which raises the partner's participation. It commits to  $x_I = 1$  when that rollout dominates the myopic choice.

Level 2 is the minimal policy that requires an agent to represent itself as an object in another agent's model; the ladder is that of the cognitive-hierarchy and level- $k$  literature [26].

#### D. Exploration

Each agent additionally participates with probability  $\epsilon$  per step irrespective of belief. This is not a numerical convenience. At  $\epsilon = 0$  the state  $p = 0$  is strictly absorbing: an agent that stops participating never observes its partner participating and therefore never revises. Escape requires both agents to explore in the same step, at rate  $\epsilon^2$ .

### III. FIXED POINTS

For a symmetric pair the participation probability obeys  $p \mapsto \mathcal{F}(p)$  with

$$\mathcal{F}(p) = \left( \left[ R_I - c \frac{1-p}{p} \right]_0^1 \right)^a. \quad (4)$$

Setting  $\mathcal{F}(p) = p$  and requiring tangency gives the saddle-node in closed form,

$$p^* = (ac)^{a/(a+1)}, \quad R_c = (p^*)^{1/a} + \frac{c}{p^*} - c. \quad (5)$$

For  $c = 0.35$  this gives  $R_c = 0.8332$  at  $p^* = 0.5916$  for  $L = 2$ , and  $R_c = 0.9818$  at  $p^* = 0.7885$  for  $L = 3$ .

Equation (5) is meaningful only while  $p^* \leq 1$ , that is

$$a \leq 1/c. \quad (6)$$

Above this bound the tangency lies outside the accessible range and there is no interior bifurcation: the coupled phase is created discontinuously at the clipping boundary  $x_I = 1$ . At  $c = 0.35$  this restricts the continuous transition to  $L \leq 3.86$ . Figure 2 shows the return map and the critical line.

Below  $R_c$  only the solitary fixed point  $p = 0$  exists. Above it there are two stable phases, coupled and solitary, separated by an unstable point whose position rises with  $a$ : more private tasks enlarge the basin of solitude. The structure is that of a stag hunt [9], with the solitary phase risk-dominant and absorbing at  $\epsilon = 0$ .

### IV. RESULTS

Unless stated otherwise, simulations use  $\tau_{\text{mem}} = 200$ , symmetric agents, and  $\geq 10^5$  steps after a 20% burn-in. We quantify the phase by the *occupancy*  $\phi$ , the fraction of time with  $p > p^*/2$ .

#### A. Validation

Reproducing Ref. [5] with i.i.d. priorities recovers  $\alpha = 1 + 1/\max_j(L_j - 1)$ . Asymmetric pairs sit on the prediction:  $(L_A, L_B) = (2, 3)$  gives  $\alpha = 1.500$  and  $(2, 5)$  gives  $\alpha = 1.250$ , both stable across four decades of the tail. Symmetric pairs approach from below —  $L = 2$  gives 1.82 at the 90th percentile, drifting to 1.94 deeper in the tail — consistent with the logarithmic correction expected when  $a_A = a_B$ . We also confirm that the mean interevent time diverges with the observation window as  $T^{2-\alpha}$ , so that for  $\alpha \leq 2$  the interaction possesses no characteristic timescale.

#### B. Modelling the partner's model shifts the critical line

Level-2 agents remain coupled well below the level-1 threshold. At  $L = 3$  a level-1 pair collapses to the solitary phase for every  $R_I \leq 0.99$ , whereas a level-2 pair remains coupled down to  $R_I = 0.90$ . The effect is markedly asymmetric in a way that inverts the  $\max_j$  rule of the non-adaptive model: at  $L = 3$  and  $R_I \in \{0.94, 0.96, 0.98\}$ , pairs of level-1 agents collapse while a level-2/level-1 pair remains coupled (Table I). In the non-adaptive model the more distracted agent sets the exponent; in the adaptive model the more sophisticated agent sets the phase.

#### C. Rare interaction requires a persistent partner model

We next decouple the rate of *opportunity* from willingness by allowing an interaction to be possible only with probability  $\nu$  per step, and by making the partner's action unobservable except when an interaction occurs. The mean gap between opportunities,  $1/\nu$ , then sets the rate at which information about the partner arrives.

If unobserved steps decay the estimate — silence treated as refusal — the coupled phase is lost whenever the estimate comes to be dominated by accumulated silence rather than by the last observed encounter. At  $1/\nu = 100$ , pairs with  $\tau_{\text{mem}} \in \{5, 50\}$  remain coupled

TABLE I. Asymmetric pairing at  $L = 3$ ,  $\tau_{\text{mem}} = 50$ . A single level-2 agent holds the pair in the coupled phase.

| $R_I$ | levels ( $A, B$ ) | phase    | occupancy |
|-------|-------------------|----------|-----------|
| 0.94  | (1,1)             | solitary | 0.000     |
| 0.94  | (2,1)             | coupled  | 0.942     |
| 0.96  | (1,1)             | solitary | 0.000     |
| 0.96  | (2,1)             | coupled  | 0.961     |
| 0.98  | (1,1)             | solitary | 0.000     |
| 0.98  | (2,1)             | coupled  | 0.980     |

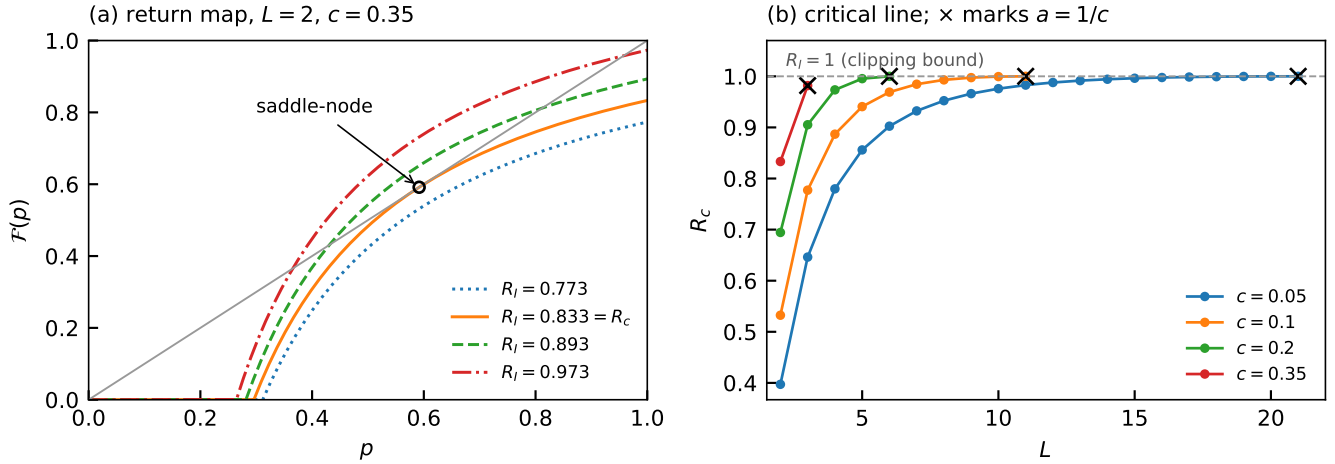


FIG. 2. (a) Return map  $\mathcal{F}(p)$  for  $L = 2$ ,  $c = 0.35$  at several  $R_I$ , with the diagonal in grey. The two nontrivial fixed points are born in a saddle-node at  $R_c = 0.8332$ ,  $p^* = 0.5916$  (circle). (b) Critical line  $R_c(L)$  from Eq. (5) for several costs  $c$ ; crosses mark the validity boundary  $a = 1/c$ , Eq. (6), beyond which no interior bifurcation exists.

TABLE II. Occupancy at  $L = 3$ ,  $R_I = 0.99$ , level 2, with the partner's action unobservable between interactions. Left: silence decays the estimate. Right: the estimate is held.

| $1/\nu$ | decaying                 |       | held                     |       |
|---------|--------------------------|-------|--------------------------|-------|
|         | $\tau_{\text{mem}} = 50$ | 5000  | $\tau_{\text{mem}} = 50$ | 5000  |
| 10      | 0.843                    | 0.000 | 1.000                    | 1.000 |
| 100     | 0.756                    | 0.000 | 1.000                    | 1.000 |
| 333     | 0.755                    | 0.000 | 1.000                    | 1.000 |

while  $\tau_{\text{mem}} \in \{500, 5000\}$  collapse.

The decisive variable, however, is not the ratio of timescales but whether the estimate is held at all. If unobserved steps leave  $\hat{p}$  unchanged, coupling survives at *every* opportunity rate and every memory constant tested, with the realised interaction rate tracking  $\nu$  to three digits (Table II). A partner model maintained through intervals containing no information is what makes rare interaction viable.

#### D. Adaptation destroys the universality classes

Away from  $R_c$  the adaptive system is not scale-free: it polarises to  $\tau = 1$  in the coupled phase and  $\tau \rightarrow \infty$  in the solitary one, with apparent tail indices in the range 4–40. Heavy tails reappear only in a narrow window above  $R_c$ , where the two fixed points nearly coincide and belief noise drives intermittent switching. Inside that window the interevent coefficient of variation reaches 8–50 and the tail is favoured over an exponential by log-likelihood differences of  $10^3$  to  $10^5$ , using the tail-comparison procedure of Ref. [8].

The window exists only for  $\epsilon > 0$ . At  $\epsilon = 0$  the solitary

phase is absorbing, the pair falls once and never returns, and no value of  $R_I$  produces a heavy tail.

Crucially, the exponent measured in the window is *not* given by  $\alpha = 1 + 1/\max_j(L_j - 1)$ . Because the transition is sharp,  $\phi$  cannot be tuned through  $R_I$ ; varying only the noise realisation at a fixed operating point, however, spreads  $\phi$  over a wide range, and  $\alpha$  tracks it (Table III). The correlation between  $\phi$  and  $\alpha$  is +0.70 at  $L = 2$  and +0.76 at  $L = 6$ ; the two  $\alpha(\phi)$  curves nearly coincide, while the non-adaptive prediction would require 2.00 and 1.20 respectively.

We conclude that  $\alpha$  is not an order parameter of the adaptive system. The occupancy  $\phi$  is the natural replacement: bounded, directly measured, and monotone in  $R_I$ ,  $L$  and modelling level, with the scale-free interevent statistics appearing as a symptom of intermediate  $\phi$  rather than as its definition.

TABLE III. Tail index versus occupancy at fixed operating points, varying only the random seed. Rows at  $\phi = 1$  have CV  $\simeq 0.76$  and are not heavy-tailed; their fitted indices are reported for completeness only.

|                                | $L = 2, c = 0.35$ | $L = 6, c = 0.10$ |
|--------------------------------|-------------------|-------------------|
| $1 + 1/\max_j(L_j - 1)$        | 2.00              | 1.20              |
| $\text{corr}(\phi, \alpha)$    | +0.70             | +0.76             |
| $\alpha$ at $\phi \simeq 0.22$ | 1.79              | —                 |
| $\alpha$ at $\phi \simeq 0.68$ | 4.59              | 4.32              |
| $\alpha$ at $\phi = 1.00$      | 4.62–4.65         | 4.37–4.42         |

### E. Bimodality, irreversibility, and the asymmetry of prevention

At fixed parameters ( $L = 2$ ,  $c = 0.35$ ,  $R_I = 0.8428$ ,  $\epsilon = 10^{-2}$ ) the outcome over 140 realisations is strongly bimodal: 45 end with  $\phi < 0.25$ , 93 with  $\phi > 0.75$ , and only 2 in between.

The outcome is nevertheless not fixed by early noise. Classifying the final phase from a trajectory prefix barely exceeds the majority-class baseline of 0.679: accuracies are 0.679, 0.693, 0.693, 0.714, 0.736 and 0.836 for prefixes of 1, 3, 10, 30, 60 and  $120 \times 10^3$  steps. The first  $10^4$  steps carry essentially no information. Collapse is a rare event available throughout the trajectory rather than a bifurcation resolved at the outset.

Collapse is, however, irreversible without intervention: none of the 45 collapsed realisations recovered when left alone. Promoting agents to level 2 at the midpoint recovers 35.6% of them, whereas pairs of level-2 agents run from  $t = 0$  on the same seeds never collapse at all (Table V). Promoting both agents is worth nothing over promoting one: recovery is driven by a single agent participating against the advice of its own estimate, and the partner need only be able to respond.

### F. The recovery cap is a limit of patience, not of reachability

The partial recovery in Table V might indicate states from which no policy in this family can return. It does not. As specified above, the level-2 rollout evaluates a single step of unreciprocated participation. We therefore introduce a commitment length  $k$ : the number of consecutive steps of unilateral participation the agent evaluates, and latches onto if it decides to proceed.

The decision is deterministic given beliefs, so the reachable region can be mapped directly. Table IV reports the smallest  $\hat{p}$  at which an agent still chooses to participate, taking  $\hat{s} = \hat{p}$ . At  $k = 1$  the agent participates only if it already believes its partner participates  $\gtrsim 46\%$  of the time — a condition a collapsed pair fails by construction. Increasing  $k$  alone is insufficient, and increasing the lookahead  $H$  alone changes nothing at all: recovery is flat at 48.5% for  $H \in \{10, \dots, 1000\}$  at  $k = 1$ . Only when a long commitment is combined with a discount horizon  $1/(1 - \gamma)$  long enough to see past it does the threshold collapse to zero, and then every collapsed realisation recovers.

The residual 48.5% is not policy-driven: it is the fraction of realisations in which exploration happens to lift  $\hat{p}$  above the participation threshold, at which point the agent latches. No state is therefore beyond recovery. What the cap measures is the reluctance of a short-horizon agent to persist without evidence. Note that the baselines in Tables V and IV (35.6% and 48.5%) come from runs of different length with different collapsed subsets and are not directly comparable.

TABLE IV. Smallest belief  $\hat{p}$  at which the agent still participates (left), and measured recovery of the collapsed realisations (right), versus commitment length  $k$  and discount  $\gamma$ . Recovery is 48.5% in every cell but one.

| $k$  | min $\hat{p}$   |                  | recovered       |                  |
|------|-----------------|------------------|-----------------|------------------|
|      | $\gamma = 0.97$ | $\gamma = 0.999$ | $\gamma = 0.97$ | $\gamma = 0.999$ |
| 1    | 0.463           | 0.458            | 48.5%           | 48.5%            |
| 10   | 0.458           | 0.453            | 48.5%           | 48.5%            |
| 100  | 0.428           | 0.313            | 48.5%           | 48.5%            |
| 1000 | 0.428           | $10^{-6}$        | 48.5%           | 100%             |

### G. Endogenous commitment: the price of recovery is a memory time

The commitment length above is imposed. We now let the agent evaluate its rollout at each candidate length and latch onto the argmax, so that the length becomes a property of its beliefs. The argmax is uninformative — it pins to the largest candidate, since once the partner participates further length is free — so we characterise the policy by the minimum patience that beats not persisting,

$$k_{\min}(\hat{p}) = \min\{k : \mathcal{R}(k) > \mathcal{R}(0)\}, \quad (7)$$

with  $\mathcal{R}(k)$  the rollout value under a  $k$ -step commitment.

Two regularities emerge, both in units of the memory constant  $\tau_{\text{mem}}$ . First,  $k_{\min}$  does not diverge as  $\hat{p} \rightarrow 0$ ; it saturates, and bisecting at  $\hat{p} = 10^{-6}$  gives  $k_{\min}/\tau_{\text{mem}} = 0.740, 0.740, 0.740, 0.738$  for  $\tau_{\text{mem}} = 50, 100, 200, 400$ . The price of recovery is therefore a fixed multiple of a memory time and is independent of how unfavourable the current estimate is. The related constant  $0.347 \tau_{\text{mem}}$  is where the partner begins to respond: its myopic priority turns positive at  $s > c/(R_I + c) = 0.293$ , reached at  $-\tau_{\text{mem}} \ln(1 - 0.293)$ . Persistence must continue about twice that long before the accumulated gain covers the commitment.

Second, whether the agent will pay that price is set by its horizon in the same units: the smallest  $H^* = 1/(1 - \gamma)$  at which an agent with  $\hat{p} = 10^{-6}$  still commits is  $H^* \simeq 1.5 \tau_{\text{mem}}$  (measured 1.50 at  $\tau_{\text{mem}} = 100, 200, 400$ ; the value 2.00 at  $\tau_{\text{mem}} = 25, 50$  is an upper bound set by the resolution of the  $H$  ladder).

These predict the simulation. At  $\tau_{\text{mem}} = 200$ , so  $H^* \simeq 300$ , an agent choosing its own commitment length recovers 48.5% of collapsed realisations at  $\gamma \leq 0.995$  ( $H \leq 200$ ) and 100% at  $\gamma = 0.999$  ( $H = 1000$ ), while the fixed  $k = 1$  control stays at 48.5% throughout. Judging recovery instead by interaction rate — an unbiased measure, since the occupancy  $\phi$  exceeds  $1/2$  whenever one agent is latched irrespective of its partner — moves the baseline to 72.7% but leaves the contrast unchanged, at 100% for the same single condition, with the mean interaction rate rising from 0.368 to 0.639 and the partner's

TABLE V. Intervention on the 45 realisations that collapsed as level-1 pairs, with promotion at the midpoint of the run.

| intervention                  | recovered | mean final $\phi$ |
|-------------------------------|-----------|-------------------|
| none (control)                | 0.0%      | 0.020             |
| agent $A \rightarrow$ level 2 | 35.6%     | 0.369             |
| both $\rightarrow$ level 2    | 35.6%     | 0.369             |
| level 2 from $t = 0$          | 100%      | 1.000             |

payoff from 0.642 to 0.734 against a solitary baseline of  $\langle x_O \rangle = 0.5$ . The recoveries are genuine couplings.

The agent that restores the interaction is not the one that gains most from it: on recovered realisations the committing agent earns 0.62–0.70 against its partner’s 0.68–0.76, the difference being the cost  $c$  of the unreciprocated offers it absorbs on the way.

## V. NETWORKS: PERCOLATION OF THE SOLITARY PHASE

The two-agent model extends to a network by giving agent  $i$  one interacting task per neighbour alongside its solitary aggregate, with a single item selected per step. Attention then becomes a scarce resource: an agent with  $k$  relations leaves  $k - 1$  offers unanswered at every step, and each relation must compete with the others as well as with the private queue. An edge executes, as before, only when both endpoints select it.

Two features of the queueing substrate turn out to be essential here and are worth stating, since abandoning either produces an artefact. First, a task’s priority must be *drawn* from a distribution the policy controls and held until execution, rather than being a deterministic function of  $\hat{p}$ . With a deterministic priority each agent locks onto its single best partner and every other relation starves, so the coupled subgraph is forced to be a perfect matching. Second, an *offer* must consume the attempt whether or not it is reciprocated. Redrawing only on execution freezes the choice: agent  $i$  offers to the same partner indefinitely, and if that partner’s best is not  $i$  the pair deadlocks with no mechanism for change.

The topology is fixed here and only the state of each edge evolves, but the coupled subgraph is itself an effective network that grows and fragments, so the setting is adjacent to the literature on adaptive networks [23], where the interplay of node dynamics with topology generates discontinuous transitions and bistability of the kind seen below.

We take the growth rules from Refs. [10, 11]: triadic closure  $LS(\ell = 1)$ , the undirected local-search rule in which a newcomer links to a random node and to one of its neighbours, closing a triangle at every step in the manner long argued to organise social ties [12]; its degree-preserving randomization; and  $BA(m = 2)$  as a non-local control. The first is local and has a finite Ramsey number

TABLE VI. Coupled edge fraction and giant component  $S$  of the coupled subgraph, at  $R_I = 0.95$ ,  $n = 600$ ,  $\langle k \rangle = 4$ , averaged over three realisations.

| $c$   | $LS(\ell = 1)$ |              | randomized |              | $BA(m = 2)$ |       |
|-------|----------------|--------------|------------|--------------|-------------|-------|
|       | edges          | $S$          | edges      | $S$          | edges       | $S$   |
| 0.002 | 0.988          | 1.000        | 0.990      | 1.000        | 0.940       | 1.000 |
| 0.010 | 0.950          | 1.000        | 0.955      | 1.000        | 0.874       | 1.000 |
| 0.020 | 0.686          | <b>0.569</b> | 0.689      | <b>0.976</b> | 0.605       | 0.967 |
| 0.030 | 0.183          | 0.006        | 0.178      | 0.006        | 0.165       | 0.007 |
| 0.070 | 0.000          | 0.000        | 0.000      | 0.000        | 0.000       | 0.000 |

$r_\kappa = 81$ , so communities are certain beyond a few hundred nodes; the other two have none. Randomizing  $LS$  preserves the degree sequence exactly, so any difference between the two is attributable to local structure alone.

We use  $c$ , the cost of an unreciprocated offer, as the control parameter, and call an edge coupled when its interaction rate exceeds ten times the solitary baseline  $\epsilon^2$ . The order parameter  $S$  is the largest connected component of the coupled subgraph as a fraction of  $n$  (Table VI).

The solitary phase percolates between  $c = 0.02$  and  $c = 0.03$ , and the interesting behaviour is at the lower edge of that window. There the two networks have statistically identical coupled edge fractions, 0.686 against 0.689, and identical degree-resolved survival, yet entirely different connectivity: the triadic-closure network has  $S = 0.569$ , thirteen components of size three or more, and a second component of 99 nodes, while its randomization remains essentially intact at  $S = 0.976$  with a second component of 4. Local structure does not determine which relations survive; it determines how their failure is arranged. Under triadic closure the coupled phase fragments into separated clusters, whereas at identical degrees without local structure it thins while staying connected. The split between local and non-local rules that governs the emergence of communities in Ref. [10] thus reappears as a split in the way the solitary phase invades.

Survival falls monotonically with degree — the fraction of a node’s relations that remain coupled at  $c = 0.020$  is 0.856, 0.704, 0.539 and 0.346 for degrees near 2, 4, 8 and 16 — and the two networks agree to within 0.01 at every degree. This follows directly from the attention constraint:  $\hat{p} \simeq P(\text{active})/k$ , so a relation is sustainable only while  $k \lesssim P(\text{active})(R_I + c)/c$ . High degree is a liability rather than a protection, because what an agent competes for is its partner’s attention, and the coincidence of the two curves shows the effect to be independent of topology.

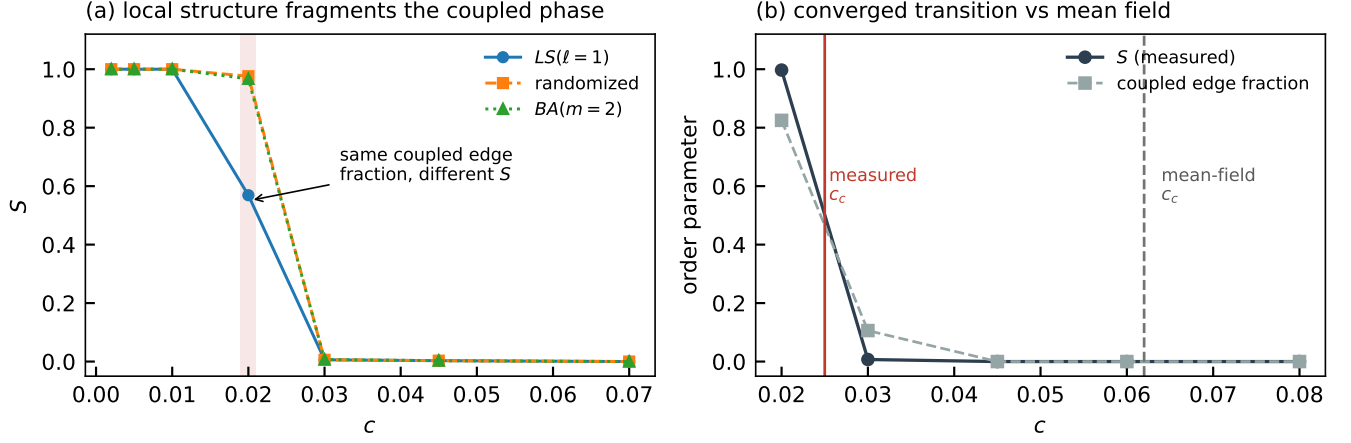


FIG. 3. Percolation of the solitary phase. (a) Giant component  $S$  of the coupled subgraph against the cost  $c$  of an unreciprocated offer, for triadic closure, its degree-preserving randomization and a non-local control ( $n = 600$ ,  $\langle k \rangle = 4$ , three realisations,  $4 \times 10^4$  steps). At  $c = 0.020$  all three carry the same coupled edge fraction but only the local rule fragments. (b) Converged runs ( $4 \times 10^5$  steps) on the randomized network place the transition between  $c = 0.020$  and  $c = 0.030$ ; the mean-field criterion of Eq. (11) predicts  $c_c \simeq 0.062$ .

#### A. Analytic threshold for arbitrary degree distribution

The transition admits a closed-form treatment. Since priorities are  $uX$  with  $u \sim U(0, 1)$  drawn per offer and the solitary task has  $F_O(t) = t^a$ , the probability that an agent of degree  $k$  selects one particular neighbour is

$$\int_0^1 u^{k-1} (uX)^a du = \frac{X^a}{k+a}, \quad (8)$$

so attention divides as  $1/(k+a)$ . Self-consistency  $X = \Psi(X^a/(k+a))$  with  $\Psi(p) = B - c/p$  and  $B = R_I + c$  gives  $X^{a+1} - BX^a + c(k+a) = 0$ , whose tangency at  $X^* = aB/(a+1)$  fixes a critical degree

$$k_c = \frac{a^a(R_I + c)^{a+1}}{c(a+1)^{a+1}} - a, \quad (9)$$

equal to  $(R_I + c)^2/4c - 1$  for  $a = 1$ . Beyond  $k_c$  no coupled state exists at any belief: attention is divided  $k$  ways,  $\hat{p} \sim 1/k$ , and it falls below the collapse threshold  $c/(R_I + c)$ .

An edge is live only if sustained from both ends, so the process is degree-targeted site percolation [13] and the Molloy–Reed criterion applies with the second moment truncated at  $k_c$ ,

$$\sum_{k \leq k_c} k(k-1)p_k > \langle k \rangle. \quad (10)$$

Equation (10) uses the full degree, but a node whose relations have already failed divides its attention only among the survivors. Writing  $\phi$  for the probability that an edge is sustained from one end and  $q_k = kp_k/\langle k \rangle$ , the cavity equation closes on a locally tree-like graph as

$$\phi = \sum_k q_k \mathbb{P}[\text{Bin}(k-1, \phi) \leq k_c - 1], \quad (11)$$

with a giant component iff  $b = \sum_k q_k \mathbb{E}[(m-1)\mathbf{1}\{m \leq k_c\}]/\phi > 1$ ,  $m = 1 + \text{Bin}(k-1, \phi)$ . Setting  $\phi = 1$  recovers (10), which is therefore an upper bound. Exploration at rate  $\epsilon$  sets  $x = 1$  and hence wins the selection, consuming the attention slot with probability  $1 - (1-\epsilon)^k$ ; including it replaces  $c(k+a)$  by  $c(k+a)(1-\epsilon)^{-k}$  in (9).

Equation (9) reproduces the measured degree dependence: survival falls monotonically with  $k$ , and  $k_c = 10.8$  at  $c = 0.020$  sits where the measured survival halves. The threshold itself, however, is a mean-field upper bound. On the randomized network, where the tree-like assumption holds, Eq. (11) gives  $c_c \simeq 0.062$  against a measured  $c_c \simeq 0.025$  — long runs place the transition sharply between  $c = 0.020$  (coupled fraction 0.825,  $S = 0.997$ ) and  $c = 0.030$  (0.106,  $S = 0.007$ ). The discrepancy is the expected one: a mean-field treatment cannot capture fluctuation-driven escape into an absorbing state, and that escape is precisely the stag-hunt mechanism, the runaway in which a falling  $\hat{p}$  lowers  $X$ , which reduces the offers received, which lowers  $\hat{p}$  again. Relations that (9) declares viable are destroyed by it.

Two practical points follow. Convergence is slow and, near the transition, directional: over  $2 \times 10^4$  to  $3.2 \times 10^5$  steps the coupled fraction *rises* from 0.488 to 0.802 at  $c = 0.020$  while *falling* from 0.165 to 0.082 at  $c = 0.030$ , so a threshold read from short runs is biased toward the initial condition. And the truncation in (10) acts at the *top* of the degree distribution. For  $p_k \sim k^{-\gamma}$  with  $2 < \gamma < 3$  the second moment diverges and the network is robust to random removal, but the removal here is targeted at high degree — the one attack to which such networks are most fragile. A scale-free contact structure is therefore maximally vulnerable to this form of collapse, for a mechanical rather than a topological reason: what an agent competes for is its partner's attention, so high

degree is a liability. The mathematics is that of intentional attack [20], but the attack is self-generated: no adversary removes hubs, and the familiar contrast between robustness to failure and fragility to targeted attack acquires an endogenous form.

The overestimate is also the expected behaviour of the approximation rather than a defect peculiar to this model. Absorbing-state transitions in noisy spatial coordination games belong to the directed-percolation class, whose upper critical dimension is four [22], and mean-field treatments systematically overestimate the extent of the ordered phase below it. We have measured no exponents and make no claim of membership in that class.

We report the transition location but claim no exponent:  $n = 600$  with three realisations is too small, and the agents here are all level 1, the network analogue of the self-model being left for future work.

## VI. GROUPS: AGENTS MEETING AS HYPEREDGES

Many joint activities are not pairwise. A meeting takes place only if everybody turns up, which makes the interacting task a hyperedge rather than an edge. We therefore let agent  $i$  hold one interacting task per hyperedge it belongs to, write  $d_i$  for that number, and require a hyperedge  $e$  of size  $m$  to execute only when all  $m$  of its members select it in the same step. Everything else is as before.

Selection is unchanged, so Eq. (8) still gives  $X^a/(d+a)$ , but an agent now needs the other  $m-1$  members rather than one, so  $\hat{p} = [X^a/(d+a)]^{m-1}$  and self-consistency yields

$$X^{n+1} - BX^n + c(d+a)^{m-1} = 0, \quad n = a(m-1), \quad (12)$$

with  $B = R_I + c$ . This has the same form as the pairwise case with  $a$  replaced by  $n$ , so the tangency argument carries over and gives a critical group degree

$$d_c(m) = \left[ \frac{n^n B^{n+1}}{c(n+1)^{n+1}} \right]^{1/(m-1)} - a, \quad (13)$$

which reduces to Eq. (9) at  $m = 2$ . The exponent  $1/(m-1)$  carries the content: the coordination requirement enters multiplicatively in the number of people who must coincide, so  $d_c$  falls very steeply with group size. At  $R_I = 0.95$  it drops from 10.76 to 1.60 to 0.67 for  $m = 2, 3, 4$  at  $c = 0.020$ , and from 112 to 6.99 to 2.51 at  $c = 0.002$ .

Because  $d_c$  can fall below one, Eq. (13) implies a *maximum group size* even for an agent that belongs to nothing else. Requiring  $d_c(m) \geq 1$  gives  $m_{\max} = 2, 3, 4, 5$  at  $c = 0.050, 0.020, 0.010, 0.002$ . It is worth noting that freely forming conversational groups are observed to cap at about four participants and to fission beyond five [24]. Two capacity arguments have been offered for that limit, one acoustic and attentional [24] and one based on a

bound on the number of other minds a participant can model at once [25]. The present mechanism is a third of the same character — attention as a scarce resource in a queue — and we make no claim that it is the operative one.

Simulation confirms Eq. (13) at  $m = 2$ : the fraction of groups that remain coupled falls from 1.000 at  $d = 1$  to 0.309 at  $d = 16$ , straddling  $d_c = 10.76$ . At  $m = 3$  it fails. Equation (13) gives  $d_c = 1.60$  and so permits  $d = 1$ , and the fixed point is genuinely there — solving (12) at  $m = 3, d = 1, c = 0.020$  gives a stable root  $X = 0.862$  against an unstable one at  $X = 0.363$ , and the simulation begins exactly on it, with a measured meeting rate of 0.14 against the predicted  $(X/2)^3 = 0.11$ . Within a few thousand steps it nevertheless collapses to zero.

The reason is that a group fails if *any single member's* belief drifts below threshold, whereupon that member withdraws, the remaining members observe no meeting, and their estimates follow. Escape from the coupled state is therefore available through  $m$  independent routes rather than one, while the coordination probability itself falls as the  $m$ th power. Both effects push the same way, and neither is visible to a mean-field treatment that asks only whether a fixed point exists. The diagnosis is confirmed by suppressing the noise: lengthening  $\tau_{\text{mem}}$ , which reduces the variance of the belief estimate, restores a nonzero meeting rate (0 at  $\tau_{\text{mem}} = 200$  rising to 0.006 at  $2 \times 10^4$  for  $c = 0.020$ , and surviving already at  $\tau_{\text{mem}} = 10^3$  for  $c = 0.005$ , where  $d_c = 4.08$  leaves a wider margin).

The rescue is only partial — the recovered rates remain an order of magnitude below the mean-field value — so the conclusion is qualitative rather than quantitative. The  $\sim 2.3$  discrepancy found in the pairwise network case becomes, for  $m \geq 3$ , a failure of kind: Eq. (13) predicts viable groups that the dynamics destroys. Group coordination is fragile not because coordinated states fail to exist, but because the absorbing state is reachable through any one participant.

### A. Reward proportional to group size

The premise so far is that a joint activity is worth the same however many take part. Often it is not: the value of the occasion grows with the number present. We therefore replace the constant  $R_I$  by  $R_I(m) = rm$  and ask whether a reward that grows with the group can offset a coordination requirement that tightens with it.

The first consequence is that the interior tangency disappears. Since  $rm > 1$  already at  $m = 2$  for any  $r$  of order unity, the optimal priority saturates at the clipping bound,  $x_j^* = 1$ , and Eq. (13) no longer applies: the coupled state is created at the boundary rather than in a saddle-node. Viability is then simply the condition that the bound be attained,  $c(1 - \hat{p})/\hat{p} \leq R_I(m) - 1$  with

$\hat{p} = [1/(d+a)]^{m-1}$ , giving

$$d_c(m) = \left[ \frac{rm - 1 + c}{c} \right]^{1/(m-1)} - a. \quad (14)$$

A numerical solution of the full fixed-point map agrees with Eq. (14) exactly, returning  $X^* = 1$  for every  $m$  it admits and  $X^* = 0$  beyond.

Equation (14) raises the ceiling substantially. At  $r = 0.95$  the maximum group size for a singly-committed agent rises from 2, 3, 4, 5 to 8, 9, 10, 13 at  $c = 0.050, 0.020, 0.010, 0.002$ . It does not, however, remove it:  $d_c \rightarrow 0$  as  $m$  grows, because the requirement  $(d+a)^{m-1} \leq (R_I(m) - 1 + c)/c$  pits an exponential against a linear function. Sustaining arbitrarily large groups would require the reward to grow at least as  $R_I(m) \gtrsim c(1+a)^{m-1}$  — exponentially in group size. No polynomial increase in the value of an occasion can compensate for the cost of getting everybody free at the same moment.

The simulations show a smaller gain still, and the discrepancy is instructive. Equation (8) was obtained assuming that the priority of each task is drawn afresh at every step. In the queueing substrate it is not: a priority is redrawn only when its task is acted on, so a membership that is seldom selected is seldom refreshed and its low priority persists. This is the same mechanism that generates heavy-tailed waiting times in the original model [1, 5], and here it depresses participation. Balancing the flux with a redraw rate  $u + \epsilon$  gives a stationary density  $\pi(u) \propto (u + \epsilon)^{-1}$ , so a singly-committed agent participates with probability

$$q = \frac{1 - \epsilon \ln[(1 + \epsilon)/\epsilon]}{\ln[(1 + \epsilon)/\epsilon]}, \quad (15)$$

equal to 0.207 at  $\epsilon = 10^{-2}$  rather than the 1/2 that a freshly drawn priority would give. The measured meeting rate at  $m = 2, d = 1$  is 0.0465 against  $q^2 = 0.0427$ , where the fresh-priority value 1/4 is wrong by a factor of five. Replacing  $\hat{p}$  by  $q^{m-1}$  in the viability condition moves the ceiling to  $m_{\max} = 4$ , and the simulations have  $m = 2$  robust,  $m = 3$  marginal at a rate of  $10^{-3}$ , and  $m \geq 4$  extinct.

The correction matters here because a singly-committed agent has only one priority to refresh. An agent holding many tasks selects the largest of them, which is refreshed often, so Eq. (8) is accurate where it was used in Secs. II–V and worst in precisely the case examined here. The conclusion is unaffected in direction: a reward proportional to group size raises the largest sustainable group but does not abolish the limit, and raises it by less than the existence criterion promises. What bounds the group is not the worth of the occasion but the rate at which everybody is simultaneously free.

## VII. CONCLUSIONS

Letting agents choose the priority of a task that requires someone else turns a stochastic process into a system with a phase transition, and the transition organises everything else.

*Its nature.* The coupled and solitary phases are separated by a saddle-node bifurcation, Eqs. (5), (9) and (13), obtained in closed form in all three settings. The transition is discontinuous: at  $R_c$  a pair of fixed points is created at finite participation  $p^*$  rather than continuously from zero, so the coupled state appears already fully formed. The solitary phase is absorbing, since an agent that stops offering never observes an offer and so never revises. Discontinuity and absorption together give the transition its asymmetry, and that asymmetry is the paper’s practical content: the conditions that sustain joint activity are not the conditions that restore it. None of our collapsed pairs recovered spontaneously; recovery required unilateral persistence for  $0.74 \tau_{\text{mem}}$ , and an agent adopts that policy only if it values the future beyond  $1.5 \tau_{\text{mem}}$ . A relation is cheap to keep and expensive to rebuild, and the price of rebuilding is set by how long it takes to be forgotten rather than by how bad things currently look.

*What it costs the established picture.* Heavy-tailed interevent times, the observation that motivates queueing models of human dynamics, are not generic once participation is chosen. Away from the transition the dynamics polarises to  $\tau = 1$  or  $\tau \rightarrow \infty$ ; scale-free statistics survive only in a narrow window near the bifurcation, and only if agents retain a nonzero rate of uninformed exploration, without which the solitary phase is strictly absorbing and no window exists at all. Within the window the measured exponent tracks the fraction of time spent coupled and not the queue lengths, so the countable family  $\alpha = 1 + 1/\max_j(L_j - 1)$  does not survive. We therefore propose phase occupancy, bounded and directly measurable, in place of  $\alpha$  as the order parameter, with scale-free interevent statistics as a symptom of sitting mid-window rather than as the object of study. Read the other way, exponents close to  $\alpha = 1 + 1/\max_j(L_j - 1)$  in correspondence data are evidence that the priority assignment there is close to exogenous; where participation is genuinely strategic, the same measurement should not reproduce them.

*Where it goes on a network.* With one shared task per neighbour, attention divides as  $1/(k+a)$  and fixes a critical degree  $k_c$  beyond which no coupled state exists. The solitary phase then percolates by the Molloy–Reed criterion with the second moment truncated at  $k_c$ , which is formally the intentional-attack calculation [20] with no attacker: scarcity removes hubs by itself. High degree is a liability, because what one competes for is the partner’s attention, and a scale-free contact structure is therefore maximally exposed to a collapse it generates internally. Local structure does not change *which* relations survive — survival is degree-controlled and indistinguishable be-

tween a triadic-closure network and its randomization — but it changes how their failure is arranged, fragmenting the coupled phase into clusters rather than thinning it uniformly. For activities requiring  $m$  participants the same construction gives a critical group degree falling as the  $(m - 1)$ th root, hence a maximum group size; there the existence criterion fails outright, because a group dies through any one of its  $m$  members while coordination falls as the  $m$ th power.

*Relevance for human dynamics.* The transition makes contact with quantities that have already been measured rather than only with exponents. A finite capacity for keeping ties active is exactly  $k_c$ , and is observed in communication records [15–17]; here it is derived rather than posited. The decay of ties whose rhythm is interrupted beyond a characteristic multiple of their own frequency [18] is the empirical form of our memory-timescale condition, and is the most promising place to test the model quantitatively. Both are statements about the coupled phase and its boundary, not about the tail of an interevent distribution, which suggests that the phase structure rather than the exponent is what the data have been describing.

*Provenance.* Several of these results — in particular the requirement that a partner model be sustained through intervals containing no observations, and the asymmetry between preventing and reversing collapse — were reached while examining a different hypothesis, that experience is a property of systems whose preferred states can be reached only through another system they do not fully control. We record this for transparency. Nothing in Secs. II–VI depends on it, and we make no claim that

the model bears on that question.

*Limitations.* The critical points are mean-field. The percolation threshold overestimates the measured one by a factor  $\simeq 2.3$ , and for group sizes  $m \geq 3$  it fails qualitatively, predicting viable groups that fluctuations destroy; both are the expected failure of a mean-field treatment of an absorbing-state transition [22], but we have measured no exponents and place the model in no universality class. Convergence near the transition is slow and directional, so thresholds read from short runs are biased toward the initial condition. The commitment length is chosen by the agent but from a fixed menu; the two recovery constants are expressed in units of a memory time that agents attribute to their partners and that all our runs took to be shared; and the network analogue of the partner-modelling agent has not been implemented.

## CODE AVAILABILITY

All code, together with a working document recording the full result log including retracted results, is available at <https://github.com/av2atgh/interagents>.

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